Robot orientation by solar polarization

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Declaration of AI Usage in the Thesis

During the preparation of this work, the author used [Tool Name, e.g., Grammarly] to check grammar, spelling, and language fluency. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of this publication.

During the preparation of this work, the author used [Tool Name, e.g., ChatGPT v4.0] to assist with structuring code snippets and brainstorming conceptual outlines. The author verified all generated outputs, critically assessed the results, and takes full responsibility for the integrity of the thesis.

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1 Introduction

Navigation has always been a primary element for survival in any species on Earth as it aids in the location of food, returning to their nesting sites, avoiding predators and migration due to climate changes. This can be from few meters to thousands of kilometres [1]. Animals use a variety of changes such as visually sited landmarks, magnetic fields, celestial bodies and skylight [2]. Other factors like birds following wind currents, fishes following different ocean currents, observing the position of the sun during different seasons in the nature to navigate vast distances without any external reliance of equipment. Humans too, have relied on environemental factors in the past and have navigated from very small distances to a difference terrain for hunting to exploring different lands thousands of kilometres away from their homes or where they were based at to also exploring different continents across the seas [3].

In the present day, we are heavily relying on the concepts and learnings of navigation to improve the manoeuvrability and the developments of robotics and drones. The advancements in the field are becoming more progressive because of the potential of including Artificial Intelligence (AI) in the pipeline. However, Navigation without a satellite fix is a persistent difficulty in mobile robotics [4]. When a robot loses GPS coverage, whether underground, inside a building, beneath a dense forest canopy, or in an area where the signal has been jammed, it must fall back on dead reckoning which enables the estimation of the robot's current orientation from previous states and in most systems this means relying on an Inertial Measurement

Unit (IMU). The accumulated error in the measurement of the acceleration and angle-rate values is initially small, so that an IMU does a good job of performance over short distances and short intervals. After some distance however, a robot needs to navigate itself through a kilometre of open area, and then this drift can start to be fast. Dead reckoning [5] uses a known position to determine the current position by continuously monitoring the distance and direction the robot has moved. If you walk across a dark room without any outside reference point (just counting steps and remembering each turn), the distance you take between steps and the angle you turn by will gradually cause the distance you think you have walked to be off by a little each time.

The same basic issue exists for an IMU. The accelerometers and gyroscopes do not sense position or heading, but rather sense changes in motion and orientation. These measurements are then summed over time, adding up small biases, noise and measurement errors [6]. This small error in heading direction can then result in a large error in the lateral position; over a long distance, it could therefore produce an inaccurate estimation of the robot's own position. It can be reduced by using an independent absolute heading reference that periodically corrects the orientation applied to dead reckoning. It is this gap that the sky-compass approach developed here aims to fill: an external reference of a heading (north-seeking direction) independent of satellite availability.

A number of alternative approaches address this problem in part. Visual odometry, wheel encoders, magnetometers, barometric altimeters, and LiDAR-based localisation each contribute something useful under the right conditions. Each also has a characteristic weakness, however, since these methods variously depend on known infrastructure, fail in featureless surroundings, or are susceptible to a particular form of interference. What would be useful is a heading reference that is passive, draws very little power, works anywhere beneath open sky irrespective of terrain, and depends on no external network at all. A sensor of this kind has existed in nature for a very long time, even though it has been slow to appear in engineering.

Desert ants of the genus Cataglyphis provide one of the most striking examples of autonomous navigation succeeding without any of these aids [7]. These small insects forage across almost featureless North African salt pans, travelling hundreds of meters from the nest in search of food and returning to it with remarkable accuracy. They make little meaningful use of landmarks. Instead, they maintain a continuously updated internal estimate of heading and distance from two sources, using step integration to measure distance and a sky compass to fix direction [8]. The sky compass reads the polarization pattern of scattered sunlight. This pattern is predictable and geometrically consistent across the whole hemisphere, and it encodes the position of the sun even when the sun itself is obscured by cloud. The eye structure responsible for reading it, the dorsal rim area, is a narrow band of photoreceptors along the top margin of each compound eye, arranged so that it samples the angle of polarisation from several sky regions at once.

The sky compass is based on the polarization pattern of scattered sunlight. This pattern is consistent and geometric throughout the hemisphere, and also describes the Sun's position when the Sun is hidden by clouds. A narrow band of specialised photoreceptors along the top of each compound eye called its dorsal rim area [9] that is necessary for reading it. This region can be considered as a narrow band of built-in polarising sensors, like a thin band of polarising sunglasses, but specialised in sensing polarised light instead of creating traditional images. The dorsal rim area samples polarization information from a number of regions at once rather than from just one direction in the sky This distributed sensing approach will enhance the solution robustness, as long as one portion of the sky is obstructed by clouds, vegetation or neighbouring objects, the rest of the sky still provides enough polarization cues to get the right orientation.

The biological mechanism is worth replicating for its tolerance rather than only for its appearance and elegance. Partial cloud cover, haze, and occlusion of the sun degrade a GPS receiver or a magnetometer-based compass well before they meaningfully affect a compass based on sky polarization. This tolerance leads to the central question of the thesis, which is whether a low-cost fisheye camera paired with a dome of polarizing film, built to follow the geometry of the insect dorsal rim area, can produce reliable heading estimates for a robot operating outdoors at high latitude.

This thesis proposes, designs, and evaluates such a system. A fisheye camera is mounted underneath a hemispherical dome of polarizing film, which is cut and assembled from gores whose transmission axes are arranged radially, so that the axis at each azimuth points away from the zenith and the sky polarization is sampled across many directions at once. An image taken through the dome records the sky polarization as a spatial pattern of intensity, and from this pattern the solar azimuth, and with it the robot's absolute heading, can be recovered. The system is validated against a solar position model developed for Turku, Finland, which computes the ground-frame azimuth and elevation of the sun for any date and time of day from astronomical parameters.

The work also has a practical motivation beyond the academic one. Autonomous outdoor platforms in Finland experience GPS degradation during geomagnetic storms, within certain urban building geometries, and in the indoor to outdoor transition zones that are common in campus and logistics settings. A passive sky compass adds an independent heading channel that complements the existing sensors rather than duplicating them, and it supplies an absolute directional reference that does not drift over time. The site itself gives a further reason to study the problem locally, since at the high latitude of Turku the summer sun sweeps through a very wide range of azimuths over the course of a day and climbs to only about 53 degrees at noon, so the sun geometry that a compass must read here differs markedly from that at the lower latitudes where most earlier systems were tested.

The remainder of the thesis is organised as follows. Chapter 2 reviews the relevant biological and engineering literature. Chapter 3 sets out the theoretical foundations, covering the solar position model, the fisheye projection, and the physics of sky polarization. Chapter 4 describes the system design. Chapters 5 and 6 cover the implementation, the experiments, and the results. Chapter 7 discusses the findings, their limitations, and directions for future work.

2 Literature Review

In this chapter the existing research that is available and serves as the basis for this thesis will be reviewed. It starts with a look at the biology of this insect sky compass, with particular emphasis on the polarization vision system of desert ants and other insects that make use of the celestial compass for navigation. The chapter then looks at the way these biological processes have been implemented in a computational model that connects biology and robotics, especially the algorithms used to estimate heading from polarized skylight. This is followed by a description of the various ways in which an artificial sky compass has been implemented, both optically and in terms of sensors, and a comparison of the proposed system and previous systems. Finally, the chapter reviews recent advances in the area of simulation environments and open-source tools, which facilitate the design, evaluation, and comparison of the different types of polarization-based navigation systems. These studies together define the state of the art, the shortcomings of existing approaches, and inspire the design choices proposed in the following chapters.

2.1 The insect sky compass

The idea that insects use the position of the sun to navigate predates modern robotics by several decades, and the mechanism behind it has come into sharper focus over time. Wehner [8] provides a thorough account of what studies of the desert ant Cataglyphis had established by that point. The ant relies on a navigational strategy built around path integration, meaning a continuously updated estimate of its position relative to the nest, computed from the direction and distance of every step taken since it set out. Direction is supplied by the sky compass, and distance by step counting. The sky compass is of particular interest because it does not read the position of the sun directly. What it reads is the polarization pattern of the sky, that is, the distribution of electric-field vector orientations across the hemisphere, which is related geometrically to the position of the sun through the physics of atmospheric scattering.

The structure responsible for this reading is the dorsal rim area (DRA) of the compound eye, a specialised band of photoreceptors along the eye's dorsal margin. The rest of the compound eye is optimised for spatial vision and colour discrimination, but in the DRA the light-sensitive microvilli are arranged in orthogonal pairs at a range of orientations. This arrangement turns each receptor pair into a polarization analyser tuned to a specific sky azimuth. The output of the DRA passes to the central complex, a brain region that has since been identified as a heading-integration circuit common to a wide range of insect species. Wehner's broader point is that Cataglyphis navigates by following a procedural routine rather than by building a map, in the sense that it maintains a heading, counts its steps, and computes the vector back to the nest. The sky compass is the anchor that keeps this heading reliable over long distances.

One detail that is easy to overlook, yet matters a great deal for an engineering implementation, is how well this strategy holds up under partial cloud cover. An insect does not require an unobstructed view of the whole sky. The polarization angle at any single point of clear sky already encodes the position of the sun, so even a small break in the clouds supplies a usable directional reference. Because the DRA integrates across space, combining many sky samples into a single heading estimate, the loss or attenuation of any individual sample does not destroy the estimate as a whole.

2.2 Computational models bridging biology and robotics

Turning the biological mechanism into a working algorithm took time, and the link between the neural circuit and the engineering solution is less direct than it first appears. The work in Gkanias et al. (2019) [10] presents a computational model that follows the known signal pathway from sky polarization input to directional output, tracing the processing stages that correspond to identified neural structures in the insect brain. The model takes the angle of polarisation measured at a set of sky azimuths and combines these values with a circular-mean algorithm that mirrors the integration carried out in the central complex. One of its more useful results is that a fairly small number of polarization samples, spread across different sky directions, is enough to recover the solar azimuth with reasonable accuracy. The practical consequence is that neither a camera with millions of pixels nor a dense array of sensors is required. A modest set of well-distributed polarization measurements suffices, and the redundancy among those measurements gives the system a natural tolerance to occlusion.

On the engineering side, Stürzl [11] presents a single-camera polarization compass with a careful treatment of measurement uncertainty. Rather than moving any component, the system images the sky through three fixed polarizers set at different orientations on one sensor, across a field of view of roughly 56 degrees, and recovers the sky polarization from these simultaneous channels. From this it computes the solar azimuth directly. Beyond the heading estimate itself, its notable feature is an explicit covariance, since the algorithm reports a quantified heading uncertainty alongside each estimate. This matters for sensor fusion. A heading measurement whose covariance is known can be weighted correctly within a Kalman filter or a comparable estimator when it is fused with IMU data, whereas without an uncertainty model the fusion depends on ad hoc tuning that may not transfer from one sky condition to another. Stürzl's work was among the first to treat the polarization compass as a true probabilistic sensor rather than a simple angle readout.

2.3 Hardware implementations and their relationship to this thesis

Moving from algorithm to physical sensor, Stürzl and Carey [12] showed that a fisheye camera system could perform polarization detection reliably enough to be carried on a UAV. The fisheye lens is a natural choice here, since a single wide-angle image captures most of the sky hemisphere, and the spatial distribution of polarization across that image carries considerably more information than any single narrow-field measurement. Their experiments showed that the solar azimuth could still be estimated reliably when the sun itself lay outside the camera's field of view, for example when the aircraft was flying with the sun behind it. This is the form of occlusion tolerance that makes the approach attractive for mobile robotics, where the orientation of the robot relative to the sun changes constantly.

The most recent and most directly relevant hardware work is that of Gkanias et al. [13], who present a physical prototype of the DRA-inspired compass. Instead of a camera, the sensor uses a ring of eight photodiode pairs, with each pair measuring two orthogonal polarization orientations. This circular layout imitates the fan-like structure of the DRA, and a circular-mean model turns the eight measurements into a solar azimuth estimate. The sensor was tested across a range of weather conditions, including heavy cloud and partial occlusion, and the spatially integrating design outperformed more computationally complex algorithms when the sky was poor. This behaviour is consistent with the biological mechanism, in that the very simplicity of the spatial averaging is what gives the system its tolerance to local disturbances in the sky.

This thesis builds on both the camera-based approach of Stürzl and Carey [12] and the bio-inspired principle of spatial integration in Gkanias et al. [13]. A fisheye camera is used in place of discrete photodiodes, which yields richer spatial information and allows the implementation to be validated against a full sky polarization model. In place of the several fixed polarizers used by those earlier camera systems, a single static dome of polarizing film is used, whose transmission axes are arranged radially, so that the whole sky polarization pattern is captured in one exposure through a single lens. What sets this design apart is not the absence of moving parts, which several of these earlier systems already achieve, but the particular combination of a single low-cost fisheye covering the whole sky hemisphere, a passive radially arranged polarizing dome, and validation at the high latitude of Turku. This distinguishes it from the narrow-field three-polarizer camera of Stürzl, the four-camera system of Stürzl and Carey, and the discrete eight-photodiode ring of Gkanias et al. The geometric design of this dome, a hemisphere assembled from gores of polarizing film, is described in detail in Chapter 4. The table 2.3 shows the previous work and the work related to this research.

2.4 Recent developments and open simulation tools

Work on the insect celestial compass has continued since the studies above. Gkanias and Webb [14] extend the compass model to the problem of time compensation, showing how a circuit of clock and compass neurons could hold a stable geocentric heading as the sun moves through the day, with the correction depending on the time of year and on the observer's latitude. The dependence on latitude is of direct interest here, since the system in this thesis is validated at a high latitude where the sun stays low and moves through a wide azimuth range. Two further studies, both in preparation at the time of writing, continue this line. Kolyfetis and colleagues [15]

System	**Sensor and polarisation sampling**	**Sky coverage**	**Test site and latitude**	**Reported accuracy**
Stürzl and Carey [12]	Four synchronised cameras with fisheye lenses and differently oriented fixed polarisers	Whole sky hemisphere	Not stated	Recovers solar azimuth and elevation; tolerant of cloud and UAV pitch and roll
Stürzl [11]	Single camera with three fixed polarisers on one sensor	Approx. 56°	Not stated	Sun direction estimated with explicit covariance
Gkanias et al. [10]	Simulated DRA-like polarisation array; model only	Simulation only	Not applicable	Proof of principle demonstrated in simulation
Gkanias et al. [13]	Ring of eight polarisation analysers, each containing four UV photodiodes at 0°, 45°, 90°, and 135°; analysers inclined at 45°	Discrete sampling; approx. 45° acceptance angle per analyser	Sardinia, Italy (39.3° N); Vryburg, South Africa (26.4° S); Bela Bela, South Africa (24.7° S)	RMSE: 5.89° global and 2.77° local for solar elevations above 10°
Current Thesis work	Single fisheye camera beneath a passive hemispherical polarising-film dome with radially arranged transmission axes	Whole sky hemisphere	Turku, Finland (60.45° N)	Reported in Chapter **??**

Table 2.1: Comparison of camera-based and photodiode-based celestial compass systems with the design presented in this thesis. Latitude is included because the geometry of the daily solar arc, and therefore the range of conditions a compass must handle, depends strongly on geographical location.

build a biologically accurate model of the polarisation vision of bees that runs from eye anatomy through to navigation, and Gkanias and colleagues [16] examine how little skylight the insect compass needs, reporting that a single ray can be enough to locate the sun. These last two are cited as directions in the field rather than as settled results, and their bibliographic details should be confirmed against the published versions before final submission.

A number of open tools now make it practical to simulate polarized skylight and to test a compass against a modelled sky rather than against measurements alone. The Prague Sky Model of Vévoda and colleagues [17] provides precomputed sky radiance across a wide spectral range and includes polarisation, and an open Python implementation of it, alongside simpler analytic sky models, is available in the sky package released by Gkanias [18]. On the measurement side, the open source OpenSky simulator of Moutenet and colleagues [19] reproduces sky polarization measurements made with a fisheye camera and compares them against outdoor captures, which is close in spirit to the validation carried out in this thesis. The present work draws on tools of this kind to render the expected polarization pattern for a given sun position and to compare the rendered pattern against captured images.

3 Theoretical Background

3.1 Solar position model for Turku

Any sky-compass system needs a way to determine where the sun should be for a given time and location [20]. Here, that calculation has two roles. It provides a ground-truth heading during the validation experiments, and it acts as a prior for the sky polarization model used in heading estimation. The solar position is computed in local ground-level coordinates at Turku, Finland, at a latitude of 60.45° N and a longitude of approximately 22.3° E.

The choice of Turku also provides a demanding test environment for a sky-compass system. At a latitude of approximately 60.45° N, the Sun remains relatively low above the horizon for much of the year, daylight duration varies substantially between seasons, and atmospheric conditions frequently include cloud cover and haze. In addition, satellite-based positioning can become less reliable at high latitudes because the geometry of visible satellites is often less favourable, reducing positioning accuracy and increasing susceptibility to signal degradation in challenging environments. These factors make it valuable to investigate an independent heading reference that does not rely on satellite signals but extract information from spatial polarization patterns [21]. A sky compass derived from the polarization pattern of scattered sunlight therefore, offers a complementary navigation cue that can continue to provide orientation information even when GNSS performance is degraded or unavailable [22].

The starting point is the Earth's annual orbit, parameterised by the fractional-year angle gamma (γ) which is denoted by:

$$\gamma = \frac{2\pi(d-1)}{365},$$

where d is the day of the year counted from 1 January. From γ, the equation of time $E(\gamma)$ accounts for the uneven orbital speed of the Earth, which arises from the ellipticity of its orbit, together with the tilt of the rotation axis. The equation gives the difference in minutes between true solar time and mean solar time, and over the course of a year it shifts true solar noon by up to about sixteen minutes relative to clock time. At Turku the mean solar noon falls at roughly 12:39 local time, and true noon departs from this by an amount that depends on the day of the year.

The solar declination δ is the angle between the equatorial plane and the direction to the Sun, and it follows the seasonal cycle set by the Earth's axial tilt of 23.44°. At the summer solstice, the Sun stands 23.44° north of the equator, while at the winter solstice, it stands 23.44° south of the equator. For a given declination δ and hour angle α, where α is measured from local solar noon at 15° per hour, the direction to the Sun can be written as a unit vector in the equatorial coordinate frame, with the z-axis pointing toward the celestial north pole.

Converting this vector to local ground-level coordinates at Turku requires a rotation matrix R built from the site latitude. In the local frame the x-axis points West, the y-axis points South, and the z-axis points vertically upward. The matrix R rotates the equatorial frame into this local frame:

$$R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \sin(\theta) & -\cos(\theta) \\ 0 & \cos(\theta) & \sin(\theta) \end{bmatrix},$$

where $\theta = 60.45°$ is the latitude of Turku. The resulting local Sun vector gives the azimuth [23], measured from South toward West, and the elevation directly. The model was implemented in Python and checked against two analytical reference values. At solar noon on the winter solstice, the elevation at Turku should equal

$$90° - 60.45° - 23.44° = 6.11°,$$

and at solar noon on the summer solstice, it should equal

$$90° - 60.45° + 23.44° = 52.99°.$$

The implementation returns 6.11° and 52.97°, respectively, which confirms its correctness to within the precision of the input constants.

The same solar geometry was implemented a second time from an independent source, using the NOAA solar position equations after Meeus, in order to guard against a hidden error in the derivation or in its port to Python. This second model agrees with the one above to a fraction of a degree across the range of dates and times of interest, which raises confidence that the ground truth used in the validation experiments is correct.

A second independent implementation was carried out from the same solar geometry using the NOAA solar position equations (Meeus) and the algorithms [24], with the result providing an independent check of both the algorithm and implementation in Python. The solar elevation at local solar noon for each day of 2026 for the NOAA implementation is plotted in Figure 3.1. Solar elevation is given by a continuous curve, which follows the normal variation according to the seasons, and by the two horizontal dashed lines representing the analytical solstice elevations (52.99° in summer, 6.11° in winter) calculated directly from the latitude of Turku and the tilt of the earth's axis without the use of any numerical model. The computed curve

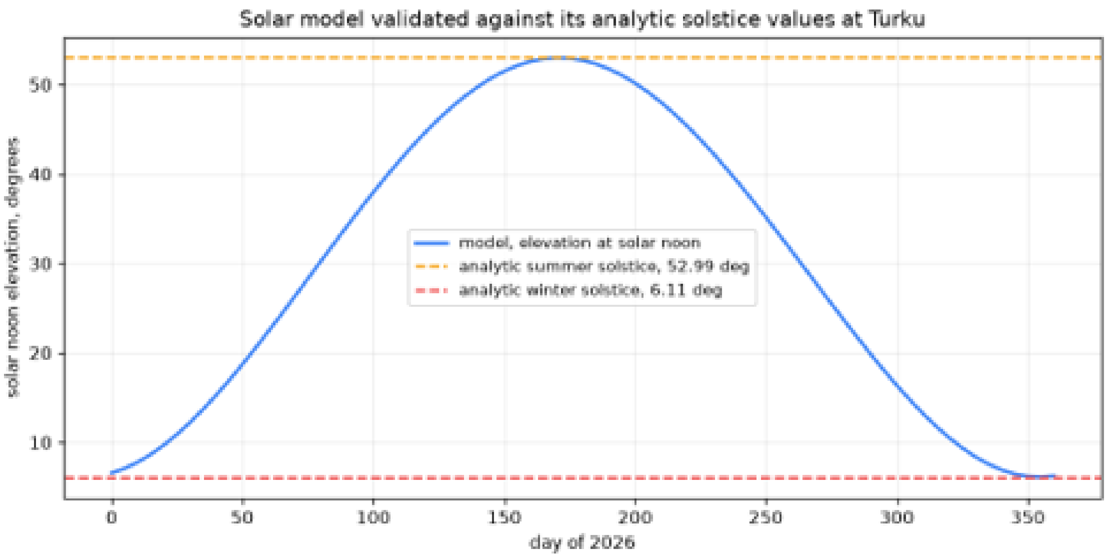

Figure 3.1: Solar noon elevation across 2026 at Turku as computed by the independent NOAA implementation, shown against the analytic solstice values for latitude 60.45 degrees North. The modelled curve meets both reference lines, with agreement within 0.01 degrees at each solstice.

crosses both the reference values on the anticipated days and the agreement is within around 0.01° around each solstice. This close agreement is much stronger evidence of correctness than a single implementation doing so because two independent derivations yield essentially identical results. Thus, there is greater confidence that the solar positions used as ground truth during the following validation experiments are accurate. This also shows an important aspect of high latitude operation, that the sunrise elevation at local noon at the latitude changes by only about 47°, from about 6° in winter to 53° in summer; and that the sun does not get near the zenith. The small number of solar elevations is a key feature of northern environments, and will be discussed in relation to high-latitude operation in Section 7.2.

One point of convention should be recorded here so that the later chapters do not silently mix frames. The derivation above measures azimuth from South toward West, following the supervisor's original formulation for Turku, and that frame is kept unchanged. The heading estimates reported from Chapter 4 onward instead use the standard robot convention, in which azimuth is measured clockwise from North. The two are related by a fixed rotation of 180 degrees, with the North referenced azimuth equal to the South referenced azimuth plus 180 degrees reduced into the range from 0 to 360 degrees, and this relation is applied wherever a solar azimuth from this model is compared against an estimated heading.

3.2 Fisheye lens projection model

The fisheye camera in this system follows the equidistant projection model, in which the pixel radius from the image centre scales linearly with the angle from the optical axis [25]. This is written as

$$\rho = a\theta,$$

where ρ is the pixel distance from the centre and θ is the angle from the optical axis, which corresponds to the zenith when the camera points straight up. The scale factor a is given by

$$a = \frac{r}{\theta_{\max}}.$$

With the camera parameters used in the supervisor's MATLAB script, $r = 470$ pixels corresponds to $\theta_{\max} = 90°$, which gives

$$a = \frac{470}{90°} \approx 5.22 \text{ pixels per degree},$$

or equivalently,

$$\frac{1}{a} \approx 0.19° \text{ per pixel}.$$

This angular resolution has a direct consequence for Sun detection. Seen from Earth, the solar disk has an angular diameter of approximately 0.53°, which, at an angular resolution of 0.19° per pixel, corresponds to a diameter of approximately

$$\frac{0.53^\circ}{0.19^\circ/\text{pixel}} \approx 2.8 \text{ pixels}$$

in the fisheye image. An object only about three pixels across, set within a sky image that is subject to lens vignetting, cloud, and bloom around bright sources, is not a reliable detection target on its own. This is the reason for adopting the polarization dome. Instead of locating a small bright disk, the system reads a broad spatial pattern that spans the whole sky image, so that the position of the sun is encoded in the structure of the pattern rather than in the location of a bright spot.

The projection model also governs the relationship between positions on the dome and directions in the sky. Each pixel in the image corresponds to a particular sky direction, and each point on the inner surface of the dome corresponds to a particular region of pixels. This mapping is used in two places. It guides the dome design, so that the polarization axes are distributed correctly across the sky azimuths, and it enters the image processing, where pixel-level polarization estimates must be referred back to sky coordinates before the solar azimuth can be computed.

The inverse of this projection is the mapping actually used when a captured image is processed. For a pixel measured from the image centre, the radial distance rho gives the angle from the optical axis as theta equal to rho divided by the scale factor a, while the direction of the pixel around the centre gives the azimuthal angle of the sky point. Together, these two angles define a unit vector pointing from the camera toward the corresponding point on the sky hemisphere. This is the same mapping the synthetic image generator uses in reverse, assigning a sky direction to every pixel so that a modelled polarization pattern can be written into the image.

3.3 Sky polarization

Before the polarization pattern is described in detail, it helps to fix a geometric frame [26]. Consider an observer at the origin O, with the zenith Z directly overhead and the sun at a position S in the sky. A viewing direction toward any point P on the sky hemisphere then makes a scattering angle gamma with the direction to the sun, where gamma is the angle at O between the line of sight to P and the line to the sun. The plane that contains both of these lines is the scattering plane for that point. The two quantities that describe the sky polarization, namely the angle of polarisation and the degree of polarisation, are both functions of this geometry, and in particular of the scattering angle gamma, so the whole pattern can be treated as a field defined over the hemisphere relative to the fixed points Z and S.

The regular polarization of skylight arises from Rayleigh scattering of direct sunlight by molecules in the atmosphere [27]. When an electromagnetic wave meets a scattering centre much smaller than its wavelength, such as a nitrogen or oxygen molecule, the scattered wave becomes partially linearly polarized. The electric field of the scattered light lies in the plane perpendicular to the scattering plane, the latter being the plane defined by the incoming ray, the scattering centre, and the observer. Across the sky this produces a polarization pattern that is symmetric about the solar meridian, the great circle that passes through the sun, the zenith, and the anti-solar point.

At any point in the sky, the Angle of Polarisation (AoP) is perpendicular to the arc that joins that point to the sun. The iso-AoP contours therefore form concentric rings around the solar position, and the whole pattern rotates as a rigid body as the sun moves across the sky [28]. The Degree of Polarisation (DoP) reaches its maximum at an angular distance of 90° from the Sun, in the plane containing the Sun and the zenith, and decreases toward zero at the solar and anti-solar points. Under clear-sky conditions, the maximum DoP is typically in the range of 70% to

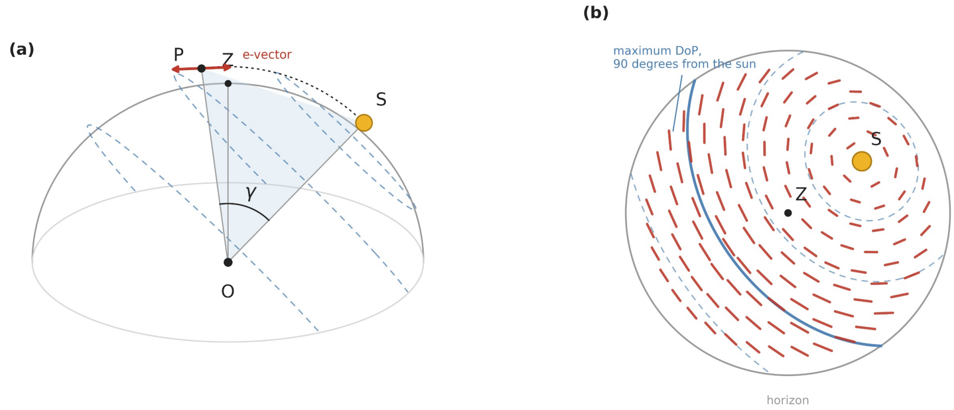

Figure 3.2: Geometry of the sky polarization pattern. (a) The observer at O, the zenith Z, the sun at S, and a sky point P. The scattering angle gamma is the angle at O between the line of sight to P and the direction to the sun, and the shaded region is the scattering plane containing both lines. The e-vector at P, shown in red, lies perpendicular to the scattering plane. (b) The same pattern in the zenith-centred projection used by the fisheye camera. Dashed curves are contours of constant angular distance from the sun, red bars give the e-vector direction with length scaled by the degree of polarisation, and the solid ring marks the band 90 degrees from the sun where the degree of polarisation reaches its maximum.

80%.

The geometry of the angle of polarisation of the sky is shown in figure 3.2 The electric field vibrates at all times at right angles to the scattering plane, and contours of constant angle of polarisation are seen as concentric rings centered on the position of the sun. The whole pattern rotates as a rigid body but maintains the same shape as the Sun travels through the sky. This predictable geometric relationship is the property that is exploited by a sky compass, with the direction of the polarisation pattern being the direction of the Sun even if the solar disk is obscured. Polarisation is also an everyday phenomenon. The atmosphere causes light from the Sun to be slightly polarised and polarising sunglasses can block glare from some parts of the sky and enhance visual contrast. The same principle is used in a sky compass that measures the orientation of the polarized light to determine heading (not reduce glare).

The degree of polarisation varies with the scattering angle in a way that Rayleigh theory makes explicit. To a good approximation, the degree of polarisation is given by

$$\mathrm{DoP}(\gamma) = d_{\max} \frac{\sin^2(\gamma)}{1 + \cos^2(\gamma)},$$

where γ is the scattering angle. The DoP falls to zero toward the Sun and the anti-solar point, where $\gamma = 0°$ or $180°$, respectively, and rises to its maximum value, $d_{\max}$, at a scattering angle of $\gamma = 90°$, along the band of sky that lies a quarter turn from the Sun.

The peak value $d_{\max}$ is determined by atmospheric conditions and typically lies between 0.7 and 0.8 under clear-sky conditions, decreasing as haze or thin cloud introduces additional unpolarised light. Since the electric field of singly scattered light lies perpendicular to the scattering plane, the angle of polarisation at each point is determined by the same geometry. This makes the polarisation pattern a rigid function of the Sun's position rather than of the local sky brightness.

The property that matters for navigation is that the AoP pattern depends mainly on the position of the sun and is far less sensitive to atmospheric turbidity, haze, or thin cloud than the sky radiance itself. The geometric relationship between the AoP and the solar position holds across a wide range of conditions. This is presumably why insects evolved a compass based on polarization rather than on intensity, since polarization carries the more reliable directional information. Gkanias et al. [5] note that the degree of polarisation and the intensity of skylight carry complementary information, and that using both signals together improves compass performance in adverse conditions.

In the dome and camera system used in this thesis, the polarizing film at each position transmits the incoming skylight according to Malus's law [29]. If the local sky has an angle of polarisation AoP and a degree of polarisation DoP, and the film transmission axis at that position makes an angle α with the reference direction, then the transmitted intensity is given by

$$I_{\text{out}} = I_{\text{in}} \left[0.5 + 0.5 \, \text{DoP} \cos \left(2(\text{AoP} - \alpha) \right) \right],$$

where I_{in} is the incident intensity. The transmitted intensity is highest where the sky polarisation aligns with the film transmission axis and lowest where the two are a quarter turn apart. Consequently, the recorded image becomes a map of how the sky angle of polarisation is oriented relative to the film transmission axis across the entire visible hemisphere. This is the model implemented in the synthetic image generator, and it is the model the heading estimator inverts.

The way the film axes are arranged across the dome therefore fixes where the low transmission band falls in the image. If a single flat filter with one fixed axis is used, the band of low transmission is set by that fixed axis and sits across the sky without regard to the sun, so it cuts across the solar position. If instead the dome is built from gores whose axes are arranged radially, so that the transmission axis at every azimuth points away from the zenith, the band of low transmission is placed so that it points toward the sun. This is the reason the radial dome is the target design rather than the flat filter, since it ties the most visible feature of the image, the dark band, directly to the quantity the system exists to estimate. The behaviour was checked by rendering the two cases with the polarization model, and the earlier description of the dome as carrying a distributed set of polarization axes should be read specifically as this radial arrangement. The intensity pattern that suits the circular mean algorithm developed from the biological model of Gkanias et al. [2] is the one produced by this radial dome.

4 System Design

This chapter describes the physical design of the sky compass and the flow of data through it. The account begins with the optical arrangement of the camera and the polarizing material, moves to the geometry of the polarizing dome and the way it is fabricated, then covers the camera and its mounting, and closes with an overview of the processing pipeline that turns a captured image into a heading estimate. The detailed methods for each processing stage are given in Chapter 5.

4.1 Optical architecture and build options

A fisheye camera points at the zenith and images the sky hemisphere through a layer of linear polarizing film. Because a single linear polarizer transmits light according to Malus's law, the intensity that reaches each pixel depends on the angle between the local sky polarisation and the transmission axis of the film in front of that part of the sky. The spatial pattern of transmitted intensity across the image therefore carries the sky polarization pattern, and with it the position of the sun.

Two build options were considered, and both are kept in the work because they trade fidelity against ease of construction. The first is a flat sheet of polarizing film held above the lens with a single fixed transmission axis across the whole field. It is simple to build and to reason about, and it serves as the first prototype for checking the capture chain and the estimation code. The second and target design is a hemispherical dome assembled from petal shaped sections of film, called gores, whose transmission axes follow a radial pattern so that the axis at each azimuth points away from the zenith. The radial dome matches the fan like sampling of the insect dorsal rim area more closely than a single flat filter, and as shown in Chapter 6 it places the low transmission band of the image in a more useful relationship to the sun.

4.2 Dome geometry and fabrication

The dome is modelled as a hemisphere of radius R covered by N identical gores. Each gore is a petal whose length from the base ring to the apex at the zenith equals a quarter of the circumference of a great circle derived by standard circle geometry [30] and is therefore given by:

$$L = \frac{\pi R}{2}.$$

The width of each gore at the base is obtained by dividing the circumference of the hemispherical base by the number of gores:

$$W = \frac{2\pi R}{N},$$

where L is the gore length, W is the gore width at the base, R is the radius of the dome, and N is the total number of identical gores.

The width of a gore at a height corresponding to the polar angle phi measured up from the base follows the same rule scaled by the cosine of phi, so each section is widest at the base and tapers to a point at the apex where all N gores meet. The prototype geometry uses R equal to 9 cm and N equal to 8, which produces gores about 14.1 cm long and about 7.1 cm wide at the base, a size that fits two gores onto a sheet of 20 by 15 cm polarizing film.

NOTE, expand in your own words (then delete this note): • Once the dome is built, record its actual measured radius and gore count here,

Figure 4.1: Image of constructed dome

and note any difference from the design figures. • This is where the outside photo of the finished dome should go, as Paavo asked, once you have taken it. [Figure 4.1: gore template, sheet layout, and assembled hemisphere from the dome cutting guide. Insert the diagrams from `dome_cutting_guide.html`**, or add the guide as an appendix.]**

Fabrication follows a printable cutting guide that provides the petal template, the sheet layout, and the assembly order. Before any film is cut, the transmission axis of the film is found with a simple crossed polarizer test and marked on each blank, because the optical function of the dome depends entirely on the axis of each gore being placed at the intended angle. The gores are then cut, their axes are checked once more against the template, the sections are joined edge to edge with a small overlap, and the apex tips are gathered and fixed at the zenith. A flat ring holds the base circle open and gives the dome a stable surface for mounting over the lens.

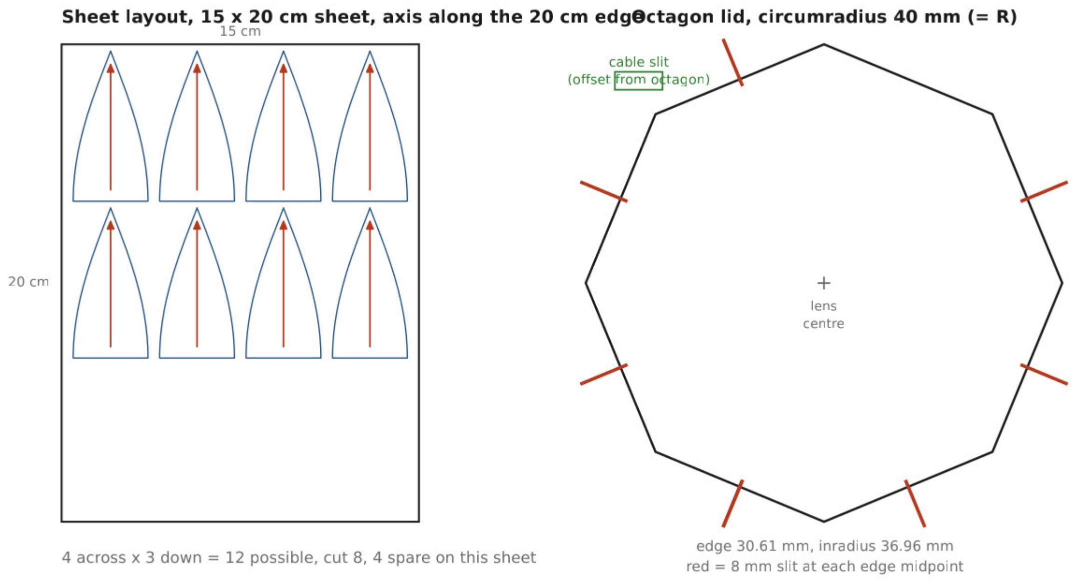

Figure 4.2: gore template, sheet layout, and assembled hemisphere from the dome cutting guide.

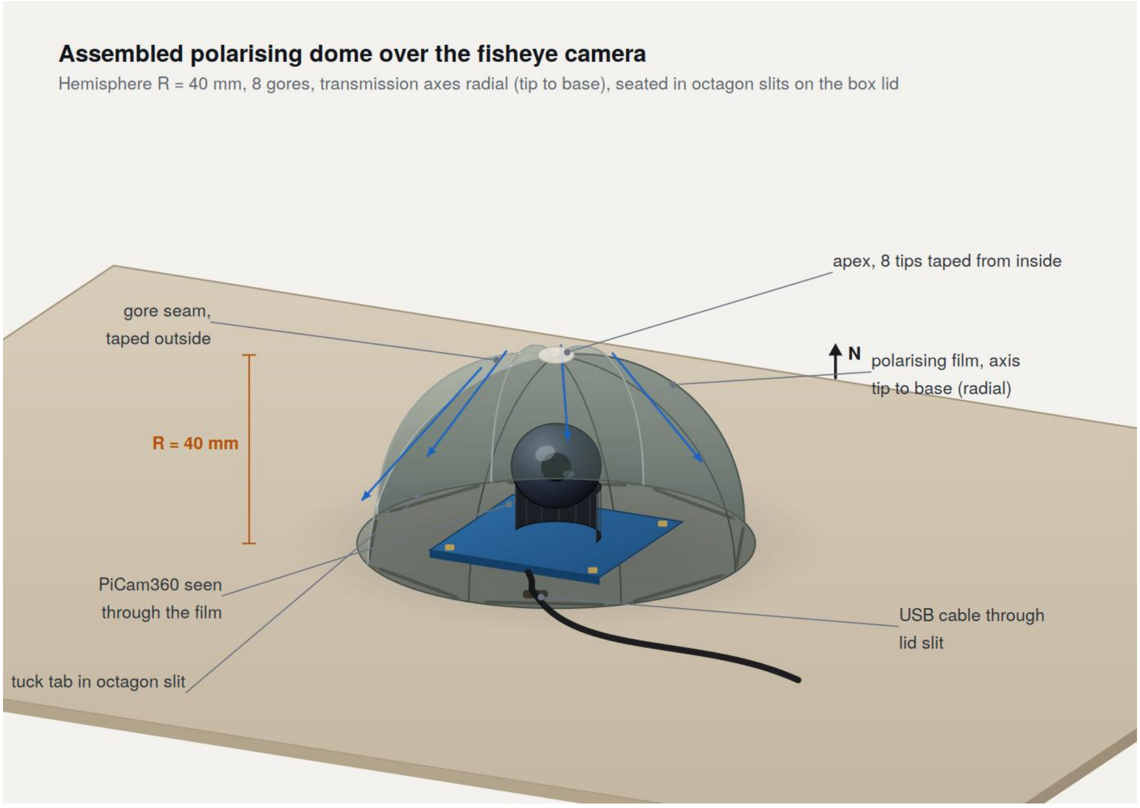

Figure 4.3: Final Dome Construction Prototype

4.3 Camera and mount

The camera is a PiCam 360 fisheye unit [31], chosen for its wide omnidirectional field with a single camera system and its support under Linux. This device is used in robotics, environmental monitoring, teleoperation, surveillance and computer vison based research [32], [33], [34]. It is mounted with its optical axis vertical so that the zenith falls at the image centre and the horizon falls on the circle at the edge of the fisheye field. The mount records the heading of a fixed reference mark on the camera body relative to the base, since the heading estimate produced by the system is expressed relative to that mark and must be tied to a known direction before it can be compared against the solar azimuth. The base is levelled so that the assumed correspondence between image radius and sky angle holds without a separate tilt correction. The radiometric settings on the camera are fixed rather than automatic, for reasons set out in the next chapter.

4.4 Processing pipeline

The path from a raw image to a heading estimate has four stages. In capture, a frame is acquired through the dome with fixed exposure and white balance. In preprocessing, the sky region is isolated from the surrounding frame and each retained pixel is associated with a sky direction through the fisheye calibration. In estimation, the observed intensity pattern is compared against a family of modelled patterns, one for each candidate heading, and the best match is selected together with a measure of confidence. In validation, the estimated azimuth is compared against the solar azimuth computed by the position model of Chapter 3 for the time and place of capture. Each stage is described in detail in Chapter 5.

[Figure 4.2: processing pipeline, from capture through preprocessing and estimation to validation.]

5 Methods

This chapter sets out the methods used at each stage of the pipeline, from configuring the camera to producing a heading estimate with an associated confidence, and it explains how labelled data are generated for development and testing.

5.1 Camera configuration

The camera runs on Ubuntu, where its controls are exposed through the Video4Linux2 interface [35]. Exposure, gain, and white balance are set to fixed values rather than being left under automatic control. This matters more here than in ordinary photography, because polarization imaging depends on the relative intensities recorded across the frame, and any automatic adjustment that changes exposure or colour balance between frames, or across regions within a frame, would alter those relative intensities and corrupt the measured pattern. Automatic white balance in particular is disabled, since it would otherwise shift the colour channels in response to the very intensity variations that carry the signal.

NOTE, expand in your own words (then delete this note): • Record the exact V4L2 values you fixed, exposure, gain, and white balance, so the capture is reproducible.

Calibration ties image coordinates to sky angles. Under the equidistant model of Chapter 3, a pixel at radius rho from the image centre corresponds to an angle theta equal to rho divided by the scale factor a, and the direction of the pixel around the

Figure 5.1: A bench test capture through the bare fisheye lens, taken at Turku on 26 July 2026 at 15:17 UTC at a resolution of 1600 by 1200 pixels in the YUYV format, with exposure, gain and white balance fixed through Video4Linux2. The polarising optic is not fitted and the optical axis is not vertical, so the frame is a test of the capture chain and of the lens geometry rather than a sky measurement. The illuminated disc has a radius of approximately 565 pixels.

centre gives the azimuthal angle of the sky point. The centre of the fisheye circle and the scale factor are found from a reference image in which the horizon ring is visible at the edge of the field, so that the mapping from pixel to sky direction is fixed before any polarization measurement is made.

The image in Figure 5.1 is a preliminary image captured with the bare fisheye lens. At this point, the polarising optic had not yet been installed, and the optical axis of the camera wasn't aligned with zenith. It is therefore not to be seen as a polarisation measurement, nor as the outcome of the sky-compass estimator. The intent of this test, however, was to confirm the entire image-acquisition chain and to look at whether the geometry of the real lens was consistent with the fisheye model used in Section 3.2.

The test results showed that the camera was recognised correctly under Ubuntu, and that photographs could be taken at the desired resolution of 1600×1200 pixels in YUYV format[36]. Manual exposure, gain and white-balance could also be done via Video4Linux2, and the images could be saved and opened up again by the processing software. The test thus confirmed the full end-to-end acquisition pathway between the physical image sensor, via the operating system camera interface, to an image that could then be processed further.

The captured frame also gives a good test of the assumed fisheye geometry. The usable image is an approximately circular area that is surrounded by a dark framed area, where the horizon is close to the edge of the fisheye image. The straighter the lines in the scene, the stronger the bend towards the image border (as in the balcony rails). These are typical of the strong radial distortion likely to be found in a fisheye lens and are consistent with employing the equidistant projection model described in Section 3.2.

Perhaps more importantly, the physical image offers an immediate estimate of the sampling angle of the camera employed in the experiments. The radius of the lighted portion of the disc measured is about 565 pixels. This radius represents the angle (90°) between the optical axis and the horizon, and results in an angular resolution of about

$$\frac{90^\circ}{565} \approx 0.159^\circ/\text{pixel}.$$

Since the apparent angular diameter of the Sun is approximately 0.53°, the solar disc would occupy only

$$\frac{0.53^\circ}{0.159^\circ/\text{pixel}} \approx 3.3 \text{ pixels}$$

across the captured image. A measurement on the same order of magnitude as the 0.19° per pixel and around 2.8-pixel solar diameter calculated in Section 3.2 using the parameters of the previous lens built by the supervisor. This is why the previous estimation is still representative, and the present estimation is based on the physical camera that was employed during the experiments.

The bench test also showed that only successful image acquisition is not enough for gaining polarisation measurement. The sky portion of Figure 5.1 is heavily saturated, with an average intensity of 220 and about 1/3 of the sky pixels being 250 or higher. If a pixel is saturated, the differences in intensity of the incident photons above the limit of the photon detector are lost and will not be recovered during further processing. Such a frame might look good for the viewer, but will not maintain the intensity variation needed for polarisation analysis. Selection of the exposure was thus considered as a calibration step in itself. The exposure time was bracketed experimentally and the longest exposure time that resulted in not saturating the relevant part of the sky was chosen for the further measurements.

The inclusion of the polarising optic also affects the correct exposure. Ideal linear polariser, by definition, will also decrease the amount of light that reaches the sensor, and in reality, any polarising film will add additional losses and may affect the colours recorded. Therefore, the exposure with the bare fisheye lens was not just scaled and used again for the finished system. Instead, the final working exposure was made with the polarising film in place so that the calibration would correspond to the optical set-up for the actual sky-compass exposures.

Another practical problem was noted when using the manual camera controls. When images were taken immediately after opening the imaging device, the brightness of the same frame can vary significantly, even if the same exposure time was used; in one test, the difference in intensity between frames taken 3 seconds apart was about 2.6. This was because the automatic camera settings were allowed to continue for a short period of time until the manual settings were applied. The acquisition process then discards a fixed number of initial frames of image data following application of the camera controls and only stores an image for analysis.

5.2 Sun detection and pixel to azimuth mapping

The sun is not found by looking for its disk. As shown in Chapter 3, the solar disk spans only about 2.8 pixels on this lens corresponding to an angular diameter of approximately 0.53° [37], which is too small and too easily confused with cloud edges and lens bloom to serve as a reliable target. The position of the sun is instead recovered from the polarization pattern that fills the whole sky image, so that the estimate draws on the entire hemisphere rather than on one small and fragile feature [28].

Once a candidate solar azimuth has been produced in the image frame, it is converted to a heading in the world frame using the fisheye calibration and the recorded camera heading. Headings are expressed in the standard robot convention [38], measured clockwise from North, which differs by a fixed rotation of 180 degrees from the South referenced azimuth used in the solar derivation of Section 3.1. The two conventions are related by setting the North referenced azimuth equal to the South referenced azimuth plus 180 degrees, reduced into the range from 0 to 360 degrees, and every heading reported in this and later chapters uses the North convention.

5.3 Orientation estimation

Heading is estimated with a model based search. For each candidate azimuth around the full circle, the expected intensity pattern that the dome would produce is rendered with the sky polarization model of Chapter 3, using the known sun elevation for the time of capture. Each rendered pattern is compared against the observed image by correlation, and the candidate whose rendered pattern correlates most strongly with the observation is taken as the estimate. The sharpness of the correlation peak provides a measure of confidence, since a sharp and isolated peak indicates a well constrained estimate while a broad or multiple peak indicates an ambiguous one [39]. A single fixed offset between the image reference mark and true North is fitted once from calibration data and applied to every subsequent estimate. The elevation of the sun is not estimated from the image but taken directly from the ephemeris, since it is already known for the time and place of capture and does not need to be recovered.

This estimator is related to the circular mean method of Gkanias et al, in that both integrate polarization information across many sky directions into a single azimuth rather than relying on any one measurement. The difference is that the method here compares the whole observed image against a rendered model of the dome, which allows the specific optical behaviour of the film and the geometry of the dome to be built into the comparison. A learned regressor trained on synthetic images is also considered, both as an alternative estimator and as a point of comparison. It maps an image directly to an azimuth without the explicit search, and its accuracy can be measured against the model based estimator on the same synthetic data.

5.4 Synthetic data generation

Labelled data are needed to develop and test the estimator before a large set of real captures exists, and ordinary photographs cannot supply them, because a normal camera records no polarization and therefore cannot show the band that the dome produces. Synthetic frames are generated instead. A numerical generator implements the sky polarization model and the film transmission law, and it renders the intensity pattern that the flat filter or the radial dome would produce for a given sun position. Each frame is labelled with the sun position that produced it, which gives an exact ground truth for training and for measuring error.

The sun positions used to drive the generator are not chosen arbitrarily. They are taken from the real record of the sun over Turku across the two months leading up to the capture campaign, sampled at ten minute steps through the daylight hours, so that the synthetic set spans the same azimuths and elevations that the system will actually meet at this latitude. Using the real history in this way is the correct use of past data for this problem, since the value of the record lies in the sun geometry it captures rather than in any polarization content, which ordinary images do not hold.

NOTE, expand in your own words (then delete this note): • Once generated, state how many synthetic frames you produced and the range of sun elevations and azimuths they cover.

6 Results

This chapter reports the results. The sun geometry that the method must handle at this latitude is described first, followed by the geometric justification for the radial dome, and then the accuracy of the estimator in simulation and against captured images. The experimental subsections are laid out here and will be completed as the capture campaign proceeds.

6.1 Sun position coverage at Turku

The range of sun positions the method must handle was taken from the solar position model for the two months leading up to the capture campaign. Across this period the azimuth of the sun spans a very wide arc, and its elevation reaches only about 53 degrees at local noon around midsummer, which is a direct consequence of the high latitude of Turku. The coverage is shown in figure 6.1.

The range of solar positions taken into account for the evaluation period are described in figure 6.1 The left-hand side displays all the sun's positions during the day for each of these dates, for example, 14 May to 14 July 2026, sampled every 10 minutes. The horizontal axis is solar azimuth (clockwise from North) and the vertical axis is solar elevation. During this time the solar azimuth ranges from about 35° to 325°, that is about 290° of the entire compass circle. The evaluation therefore tests the estimator with a wide range of headings that it could potentially see in its out-of-doors operation, and not just a small sample of solar directions.

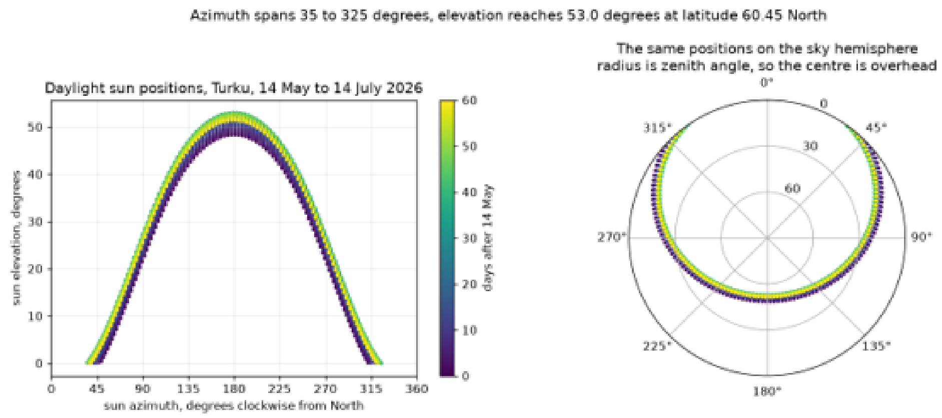

Figure 6.1: **Daylight solar positions at Turku between 14 May and 14 July 2026, sampled at ten minute intervals and coloured by day. Azimuth spans approximately 35 to 325 degrees and elevation reaches 53.0 degrees, which is the range of sun geometry the compass is required to handle at latitude 60.45 degrees North. The right panel shows the same positions projected onto the sky hemisphere, where the radius is the zenith angle, so the centre of the plot is directly overhead.**

The highest point that the sun ascends during the time interval is about 53.0°. The Sun's elevation angle is thus, as a direct consequence of the latitude of Turku 60.45° N, more than 37° away from the zenith even at the summer solstice. The colour scale indicates the number of days since 14 May, and the seasonal evolution can be seen. As the summer solstice nears the solar arcs grow further and further out, and the peak days of high elevation occur during the solstice itself, after which they start to recede and the solar arcs gradually shrink. The figure thus not only reflects the various directions of the sun during the course of a day, but also the different solar geometries over the two-month period under consideration.

The right side shows the same solar positions displayed as a hemispherical sky. The projections in this projection display the zenith angle on the radial coordinate; the centre of the chart is the zenith and the further out, the closer to the horizon. The resulting solar trajectories form a broad band away from the centre of the hemisphere. This gives a direct visualization of the high-latitude solar geometry; the Sun is well off the zenith and its elevation angle stays mostly in the fringe of the visible hemisphere of the sky in the course of the whole evaluation period.

The total number of daylight solar positions calculated over the evaluation window is 5972. The test set for evaluating the estimator in Section 6.3 was a random sample of 60 of this population. Such samples of all daylight positions enable an evaluation to be conducted for a variety of azimuths, elevations, times of day and dates, and not just for a few hand picked solar positions.

The relatively low solar elevation also had some practical implications for outdoor image capture. A maximum elevation of some 53° indicates that the Sun never rises high above the horizon in the summer time, even in the most clear and sunny summer days in Finland. In reality, this results in longer shadows than at lower latitudes and makes the brightest part of the sky further away from the centre point of the image and closer to the horizon in a zenith facing fisheye shot. This is especially the case for exposure and dynamic range, as the bright areas near the horizon can compete with much darker areas of the sky. This experience showed that it is important to test the sky-compass system in the real solar geometry of the operating environment, rather than by making assumptions for lower latitudes.

6.2 Dome band geometry

The geometric argument for the radial dome is shown by rendering the intensity pattern that each build option produces. A single flat filter with a fixed transmission axis places a band of low transmission across the sky, oriented by the fixed axis rather than by the sun, so the band sits across the solar position. A radial dome, whose transmission axis points away from the zenith at every azimuth, instead places the band so that it points toward the sun, which ties the most visible feature of the image directly to the quantity being estimated. The two geometries are compared in figure 6.2, and a single captured time rendered as a synthetic frame is shown in figure 6.3.

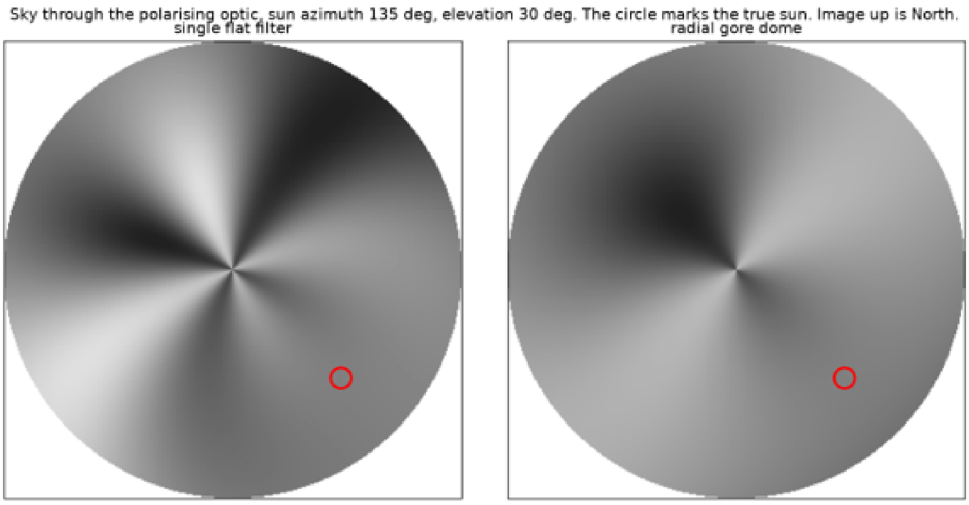

Figure 6.2: **Modelled sky radiance through the polarising optic for a sun azimuth of 135 degrees and an elevation of 30 degrees, with image up corresponding to North. The left panel is the single flat filter and the right panel is the radial gore arrangement. The red circle marks the true solar position in both.**

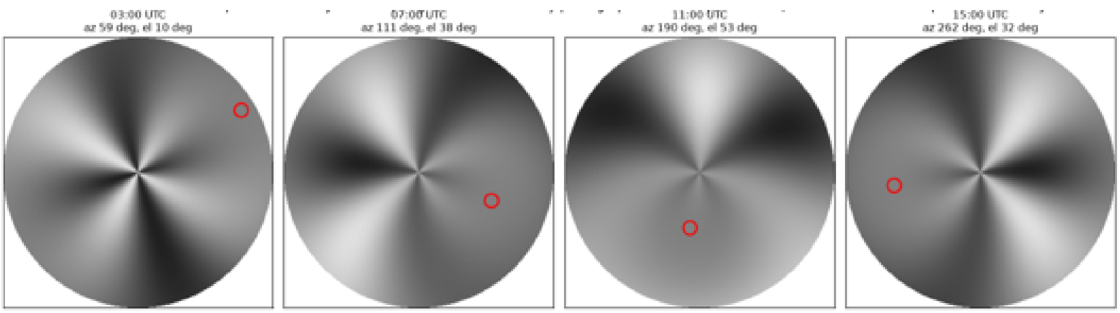

Figure 6.3: **Synthetic frames rendered through the flat filter at the true solar positions for 21 June 2026 at Turku. The solar azimuth and elevation used to render each frame are given above it and the red circle marks the resulting sun position in the image. Because ordinary photographs record no polarisation information, labelled frames for development and testing are produced in this manner from measured solar geometry.**

NEW. Points on the content of Figures 6.2 and 6.3, to expand into the paragraphs above (then delete this note): • Figure 6.2: both panels are rendered for the same sun, azimuth 135 degrees and elevation 30 degrees, so any difference between them comes from the optic alone and not from the sky. • Figure 6.2: the four lobed pinwheel structure in both panels comes from the factor cos(2(AoP minus alpha)) in Malus's law, since the transmitted intensity completes two full cycles as the analyser angle turns once through the circle. • Figure 6.2: the flat filter produces markedly higher contrast between its light and dark lobes, while the radial arrangement produces a smoother and flatter field. • CHECK BEFORE YOU WRITE: the existing prose in 6.2 states that the radial dome places the dark band so that it points toward the sun. Look carefully at the right panel against the red circle and confirm that the rendering actually shows this. If what the figure shows is a difference of phase and contrast rather than a band that points at the sun, the sentence must be softened to match the figure. Do not leave a claim in the text that the reader can see is not in the picture. • Figure 6.3: the four frames are 03:00, 07:00, 11:00 and 15:00 UTC on 21 June 2026, with the sun at azimuth 59, 111, 190 and 262 degrees and elevation 10, 38, 53 and 32 degrees respectively. • Figure 6.3: the red circle moves clockwise around the image and inward toward the centre as the elevation rises, reaching its closest approach at 11:00 when the sun is at 53 degrees, then moving outward again. This is the equidistant projection of Section 3.2 doing exactly what it should. • Figure 6.3: the whole intensity pattern rotates rigidly with the sun rather than changing shape, and that rigidity is the property the estimator exploits when it searches over candidate azimuths. • Figure 6.3: the 11:00 frame is visibly flatter than the others. Note that pattern contrast

falls as the sun climbs, which is one reason the elevation is supplied from the ephemeris rather than estimated from the image.

6.3 Estimator accuracy in simulation

This section reports the azimuth error of the estimator on synthetic frames across the full range of sun positions covered in Section 6.1. Because the synthetic frames carry an exact ground truth, the error here isolates the behaviour of the estimator itself from any error in capture or calibration.

The estimator was evaluated on 60 solar positions drawn at random from the 5972 real daylight positions of Section 6.1, under four conditions: a clear sky through the flat filter, thin cloud, heavy cloud, and a clear sky through the radial gore dome. The results are collected in Table 6.1.

Table 6.1: **Absolute azimuth error in degrees over 60 solar positions drawn at random from 5972 real daylight positions at Turku between 14 May and 14 July 2026. The film axis was 35 degrees and the image up heading 0 degrees throughout, with sensor noise of 0.020 in all four runs.**

Condition	**Optic**	**d_{max}**	**Clutter**	**Mean**	**Median**	**P90**	**Max. err.**	**Within 5°**	**Conf.**
Clear	Flat filter	0.75	0.00	0.075	0.073	0.144	0.206	100.0%	1.78
Thin cloud	Flat filter	0.45	0.30	0.984	0.671	2.252	4.125	100.0%	1.78
Heavy cloud	Flat filter	0.20	0.60	16.259	3.254	14.519	179.657	60.0%	1.74
Clear	Radial dome	0.75	0.00	0.075	0.074	0.130	0.224	100.0%	1.76

7 Discussion, Limitations, and Future Work

This chapter reflects on what the design achieves, the conditions under which it is expected to work, and the directions in which it could be extended.

7.1 Limitations

Several limitations follow directly from the design and should be stated plainly. The first concerns the use of synthetic data. Because an ordinary photograph records no polarization, historical real imagery cannot show the band that the dome produces, and synthetic frames rendered at real sun positions are used in its place for development and for the simulation results. This is a reasonable substitute, but it means that the simulation accuracy reflects the fidelity of the model rather than the behaviour of the physical dome, and only the captured image validation of Section 6.4 tests the latter.

A single polarizing film also leaves a solar and antisolar ambiguity, because the transmission pattern is symmetric under a rotation that exchanges the sun with the point opposite it in the sky, and the estimator must resolve this ambiguity rather than assume it away. The method further depends on a reasonably clear view of some part of the sky, since it needs polarized skylight to read, and although it tolerates broken cloud far better than a sun tracker would, a fully overcast sky with no polarization structure leaves it without a signal. Finally, any error in the recorded camera heading maps directly onto the reported azimuth, so registration of the camera to a known direction is as important as the estimation itself, and the flat filter prototype, being lower in fidelity than the radial dome, is expected to give correspondingly weaker estimates.

NEW. Two limitations to add to the paragraph above, in your own words (then delete this note): • The confidence metric does not discriminate. The mean confidence reported in Table 6.1 varies by less than three percent across conditions whose mean error differs by more than two orders of magnitude, so the sharpness of the correlation peak, as currently defined, does not distinguish a reliable estimate from a failed one. This matters because the intended use of the confidence value was to weight the heading within a fusion filter and to reject the solar and antisolar failures, and neither is possible until the metric is redefined. Frame this as a known and identified weakness with a clear route forward, since an examiner will respect that far more than silence. • The synthetic and the captured validations answer different questions, and this thesis answers the first thoroughly and the second partially. The simulation establishes internal consistency of the geometry, the search and the coordinate conventions; only captured frames test whether the forward model resembles the real sky. Frame the gap as scheduled rather than overlooked.

7.2 High latitude considerations

The site places particular demands on the method. At the latitude of Turku the summer sun reaches only about 53 degrees above the horizon at noon and sweeps through a very wide range of azimuths over the course of a day, so the estimator is exercised across almost the whole azimuth circle but never sees a sun near the zenith. This differs from the middle and lower latitude conditions under which much of the earlier work was carried out, and it is one reason the local sun geometry is worth validating against directly. A useful extension would repeat the experiments in October, when the sun is lower and follows a shorter and differently shaped daily arc, so that the behaviour of the method can be checked under a second and quite different sun geometry at the same site.

NOTE, expand in your own words (then delete this note): • Explain in a sentence why October changes the arc, since the sun rises lower, travels a shorter path, and sets sooner than at midsummer. • Note how you plan to capture the sun then, for example the snapshots and the annual movement video.

7.3 Future work

Several extensions follow from this work. The capture campaign could be repeated in October as noted above, and extended to collect data during rain, so that the effect of rain on the polarization pattern can be observed rather than assumed. On the application side, a passive heading reference of this kind is of interest for environmental monitoring and for other outdoor observation tasks where a robot must hold a heading without a satellite fix.

On the method side, a tilt correction would remove the requirement for a level base and could follow the approach taken in the hardware compass of Gkanias et al. [5], and fusion with an inertial measurement unit would let the absolute heading from the sky compass bound the drift of the inertial estimate, using the covariance idea of Stürzl [3] to weight the two sources correctly. A learned estimator trained on a larger synthetic and real dataset could be compared against the model based estimator, and the whole system could be cross checked against measured sky polarization datasets and against the recent insect compass model of Gkanias and Webb [6] together with the related work now in preparation.

NOTE, expand in your own words (then delete this note): • Explain briefly why rain interferes, since water on the film and droplets in the air scatter and depolarise the light, and frame this as making the dome more versatile for poor weather.

8 Conclusion

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Appendix A Full Weekly Results